

Saanidhya Shahi
09619051623
AIML B2
Fuzzy Logic

Exp. 3 : To implement and analyze Yager fuzzy intersection, union, and various fuzzy complement operators using MATLAB.

1. Construct:

- i) How does the parameter w affect the shape and behavior of the Yager fuzzy intersection surface?
- ii) Does the Yager intersection converge to $\min(a,b)$ as $w \rightarrow \infty$?

Code :-

```
a = 0:0.05:1;
b = 0:0.05:1;

[A,B] = meshgrid(a,b);
w_values = [0.5 1 2 5];

figure;
hold on;

for k = 1:length(w_values)

    w = w_values(k);

    S = (1 - A).^w + (1 - B).^w;
    S = min(1,S);
    I = 1 - (S.^(1/w));

    surf(A,B,I,'FaceAlpha',0.6)
```

End

```

xlabel('a'); ylabel('b');
zlabel('I_w(a,b)') title('Yager
Intersection (Different w)')
legend('w=0.5','w=1','w=2','w
=5')

```

```

grid on
view(135,30)

```

2:

```

a = 0:0.05:1;
b = 0:0.05:1;

```

```

[A,B] = meshgrid(a,b);

```

```

w_large = 1000;

```

```

S = (1 - A).^w_large + (1 - B).^w_large;
S = min(1,S);
I_large = 1 - (S.^(1/w_large));
I_min = min(A,B);

```

```

figure; hold on;
surf(A,B,I_large,'FaceAlpha'
,0.7)
surf(A,B,I_min,'FaceAlpha',
0.7) xlabel('a'); ylabel('b');
zlabel('I(a,b)')
title('Intersection:  $w \rightarrow \infty$  vs
min(a,b)')
legend('w=1000','min(a,b)')

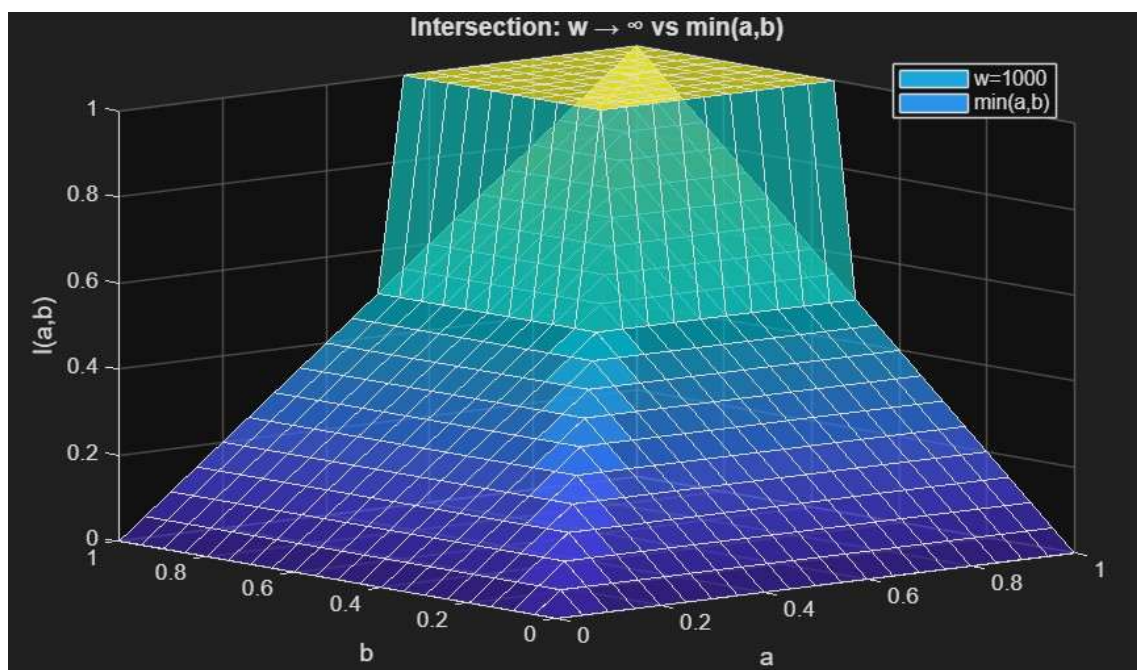
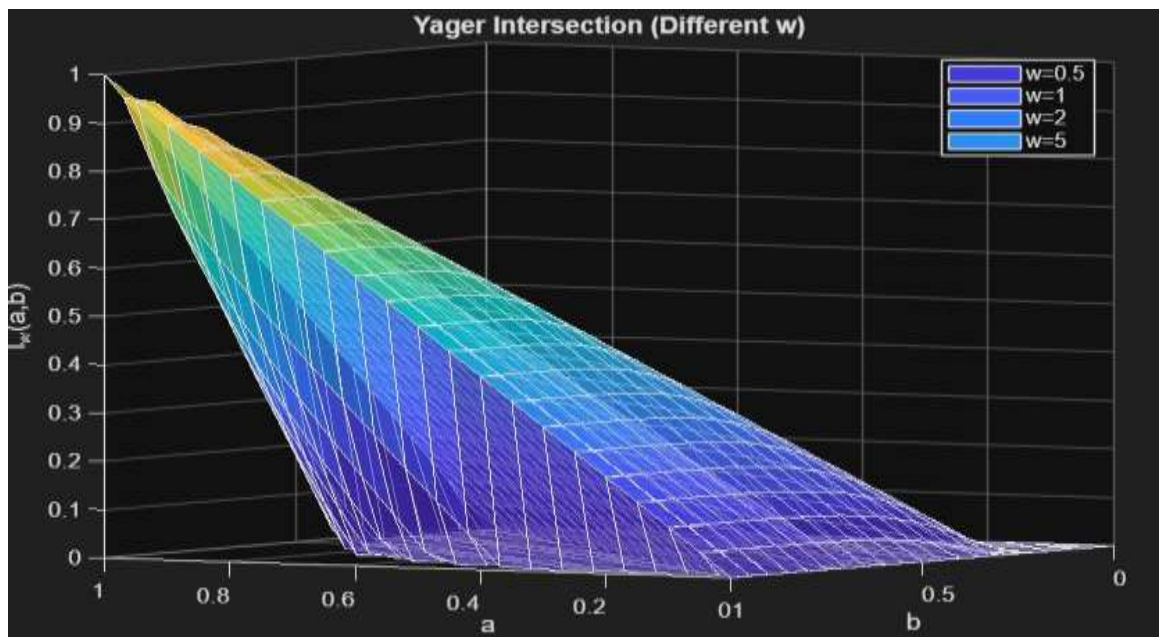
```

```

grid on
view(135,30)

```

Output:



2. Construct:

- i) How does varying w influence the Yager fuzzy union surface?
- ii) Does the Yager union converge to $\max(a,b)$ as $w \rightarrow \infty$?

Code:-

```
a = 0:0.05:1;  
b = 0:0.05:1;
```

```
[A,B] = meshgrid(a,b);  
w_values = [1 2 5 20];
```

Figure;

```
hold on; for k =  
1:length(w_values)  
  
    w = w_values(k);  
  
    U = (A.^w + B.^w).^(1/w);  
    U = min(1,U);  
  
    surf(A,B,U,'FaceAlpha',0.6)  
  
end  
xlabel('a'); ylabel('b');  
zlabel('U_w(a,b)') title('Yager  
Union (Different w)')  
legend('w=1','w=2','w=5','w=20')  
grid on  
view(135,30)
```

2:

```
a = 0:0.05:1;
```

```
b = 0:0.05:1;
```

```
[A,B] = meshgrid(a,b);
```

```
w_large = 1000;
```

```
U_large = (A.^w_large + B.^w_large).^(1/w_large);
```

```
U_large = min(1,U_large);
```

```
U_max = max(A,B);
```

Figure;

```
hold on;
```

```
surf(A,B,U_large,'FaceAlpha',0.7)
```

```
surf(A,B,U_max,'FaceAlpha',0.7)
```

```
xlabel('a'); ylabel('b');
```

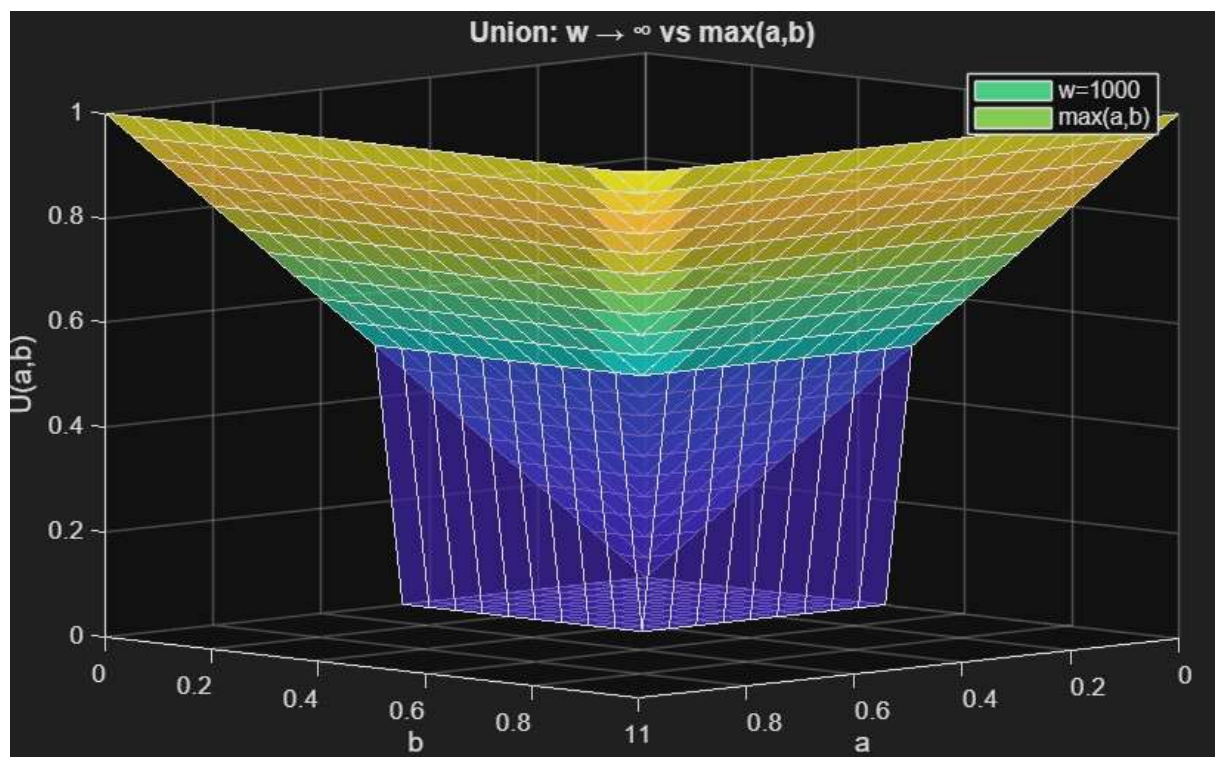
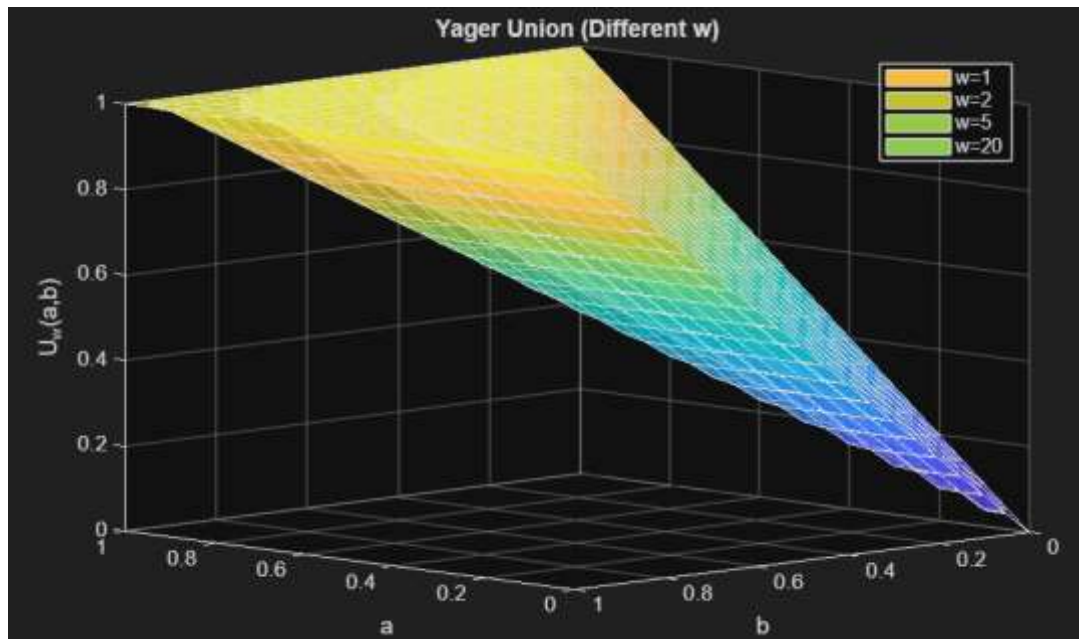
```
zlabel('U(a,b)') title('Union: w  
→ ∞ vs max(a,b)')
```

```
legend('w=1000','max(a,b)')
```

```
grid on
```

```
view(135,30)
```

Output:



3. How do different fuzzy complement operators differ in behavior, and how does parameter variation affect them?

Code:

```
a = 0:0.01:1;
```

```
[X,Y] = meshgrid(a,a);
```

```
C1 = 1 - X;
```

```
C2 = 0.5*(1 + cos(pi*X));
```

```
T = Y;
```

```
C3 = double(X <= T);
```

```
w = 0.1 + 5*Y;
```

```
C4 = (1 - X.^w).^(1./w);
```

Figure;

```
subplot(2,2,1)
```

```
surf(X,Y,C1)
```

```
shading interp
```

```
title('C(a)=1-a')
```

```
xlabel('a')
```

```
ylabel('parameter')
```

```
zlabel('C(a)')
```

```
view(135,30)
```

```
subplot(2,2,2)
```

```
surf(X,Y,C2) shading
```

```
interp
```

```
title('C(a)=0.5(1+cos(
```

```
\pi a))' xlabel('a')  
ylabel('parameter')  
zlabel('C(a)')  
view(135,30)
```

```
subplot(2,2,3)  
surf(X,Y,C3) shading  
interp title('Threshold  
Complement')  
xlabel('a') ylabel('t')  
zlabel('C(a)')  
view(135,30)
```

```
subplot(2,2,4)  
surf(X,Y,C4)  
shading interp  
title('Yager  
Complement')  
xlabel('a')  
ylabel('w')  
zlabel('C(a)')  
view(135,30)
```

```
grid on
```


Output:

